

ENVY-FREE CHORE ALLOCATION WITH UNIT SUBSIDIES UNDER NEGATIVE DICHOTOMOUS VALUATIONS

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Abstract

We study the allocation of indivisible chores with subsidies under negative dichotomous valuations, where the marginal disutility of every chore is either zero or one. This is the chore analogue of the dichotomous goods model of Barman et al. [BKNS22], for which a subsidy of at most one unit per agent is known to suffice for envy-freeness.

We show that the same optimal guarantee holds for chores, under no structural assumption beyond binary marginals. For every instance with negative dichotomous valuations, there exists a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with $\sum_{i \in N} p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is envy-free. Moreover, the allocation $\mathcal{A}$ is EF1 even before subsidies are paid. The bound is tight, and the allocation and subsidies can be computed in polynomial time in the value-oracle model.

Our algorithm starts from an envy-free partial allocation produced by the algorithm of Tao et al. [TWYZ25], assigns the remaining chores to distinct agents in a tail strongly connected component selected at termination, and determines the subsidised agents through a backward closure in the associated equality graph.

1 Introduction

Discrete fair division studies the allocation of indivisible resources among agents with heterogeneous preferences, and lies at the interface of mathematical economics and computer science [BCE⁺16, End17]. A central fairness criterion is *envy-freeness* [Fol67, Var74]: no agent prefers another agent's bundle to her own. For indivisible items, however, an envy-free allocation need not exist, as the standard one-item two-agent instance illustrates. This has motivated a large literature on relaxations of envy-freeness, including envy-freeness up to one good (EF1) [Bud11, LMMS04] and the stronger notion of envy-freeness up to any good (EFX) [CKM⁺19].

A second and complementary line of work retains exact envy-freeness by allowing monetary subsidies. In addition to her allocated bundle A_i , each agent i receives a subsidy $p_i \geq 0$ from a third party, and the outcome is envy-free if

$$v_i(A_i) + p_i \geq v_i(A_j) + p_j \quad \text{for all } i, j.$$

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Halpern and Shah [HS19] show that every instance admits an envy-freeable allocation and study the minimum subsidy required to achieve envy-freeness. Under the standard normalization that each item value is at most one,¹ they show that, for a given envy-freeable allocation, a total subsidy of at most $(n - 1)m$ may be required in the worst case, and conjecture that a total subsidy of $n - 1$ always suffices when the allocation can also be chosen. Brustle et al. [BDN⁺20] prove this conjecture for additive valuations, obtaining an allocation that is also EF1 and balanced, and show that for general monotone valuations a subsidy of at most $2(n - 1)$ per agent suffices.

Barman et al. [BKNS22] obtain a substantially stronger guarantee for dichotomous valuations. When every marginal $v_i(S \cup \{g\}) - v_i(S)$ belongs to $\{0, 1\}$, they show that there exists an allocation admitting subsidies in $\{0, 1\}^n$, with total subsidy at most $n - 1$, and that such an outcome can be computed in polynomial time in the value-oracle model. This bound is tight: for n agents and a single good valued at one by every agent, the $n - 1$ agents who do not receive the good each require one unit of subsidy.

In this paper, we consider the corresponding problem for indivisible chores, where receiving an additional item weakly decreases an agent’s utility. Chore-allocation problems arise naturally in applications such as shift assignment, on-call duties, teaching and refereeing loads, committee service, and maintenance tasks. Monetary compensation is also natural in such settings, where it may be interpreted as compensation for undertaking an undesirable task or additional workload.

We study *negative dichotomous* valuations: for every agent i , every $S \subseteq M$, and every $g \in M \setminus S$,

$$v_i(S \cup \{g\}) - v_i(S) \in \{0, -1\}.$$

Thus, every item is a chore for every agent, and adding a chore changes an agent’s utility by either zero or minus one. This domain is the sign analogue of the dichotomous goods model of [BKNS22].

Binary marginals capture settings in which an additional task either creates a new unit of burden or can be absorbed by an existing commitment. For example, an agent already covering an on-call window may incur no additional cost from another incident during that window, or a driver already routed through a district may absorb an additional stop at no extra cost. Since whether a chore is costly may depend on the rest of the assigned bundle, the model naturally allows non-additive valuations, including valuations that are not subadditive.

The baseline and the lower bound. Two observations provide useful benchmarks for the problem, and we state them explicitly so that the contribution can be measured against them. On the one hand, the existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS⁺25, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies directly and yields a subsidy of at most $n - 1$ per agent and $n(n - 1)/2$ in total [KMS⁺25, Theorem 1]. Two details concerning this application are worth noting. Their normalisation is two-sided, so it

¹Such a normalization is necessary when stating absolute bounds on monetary subsidies.

bounds chore disutility as well as value; and the EF1 they require is the two-sided form of Definition 5, which removes one item from *either* bundle, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS⁺25, §2] cites [LMMS04] and [BSV21]. In particular, while [LMMS04] concerns goods, [BSV21] provides the corresponding existence result required in the chore setting.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n - 1$ lower bound transfers). Take n agents and $m = n - 1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n - 1$ agents hold one chore each and one agent holds nothing, each of the $n - 1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n - 1$. No allocation does better: consider any complete allocation $\mathcal{A} = (A_1, \dots, A_n)$, and write $s_i = |A_i|$. Since there are only $n - 1$ chores, $\sum_{i=1}^n s_i = n - 1$, and therefore at least one agent, say h , receives no chore: $s_h = 0$. Suppose that a subsidy vector $p \in \mathbb{R}_+^n$ makes the allocation envy-free. In cost form, envy-freeness requires

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j.$$

Comparing every agent i with the empty-bundle agent h , we obtain

$$s_i - p_i = c_i(A_i) - p_i \leq c_i(A_h) - p_h = -p_h.$$

Hence

$$p_i \geq s_i + p_h \geq s_i \quad \text{for every } i.$$

Summing over all agents gives

$$\sum_{i=1}^n p_i \geq \sum_{i=1}^n s_i = n - 1.$$

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Theorem 2 (Main result). *For every instance with negative dichotomous valuations, there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0, 1\}^n$, hence with total subsidy at most $n - 1$; the allocation $\mathcal{A}$ is moreover EF1 on its own, before any subsidy is paid; and such an allocation and subsidy vector can be computed in polynomial time given value-oracle access.*

Example 1 shows this bound is tight. The guarantee of [KMS⁺25] is $n - 1$ per agent and $n(n - 1)/2$ in total, whereas our target is at most one per agent and $n - 1$ in total, yielding a linear-factor improvement.

Our contribution. We establish the statement above, for every number of agents (Theorem 22). The allocation produced is EF1 in the sense of Definition 6 before any money changes hands, and the addition of subsidies upgrades the resulting allocation to exact envy-freeness. The proof takes as input an envy-free *partial* allocation — the algorithm of Tao et al. [TWYZ25] produces one in polynomial time, leaving at most $n - 1$ chores unassigned — and completes it: the residual chores are assigned to distinct agents inside the strongly connected component at which that algorithm halts, and the paid set is the backward closure of those recipients under the equality relation $c_i(X_i) = c_i(X_j)$. Envy-freeness of the completion is verified directly, so no appeal to the characterisation of envy-freeable allocations is required. Section 2 fixes the model and records what transfers from the goods theory unchanged; Section 3 contains the proof.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n - 1$, Goko et al. [GIK⁺24] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The work of Kawase et al. [KMS⁺25] is particularly relevant to the present setting because it handles general doubly monotone valuations, and hence includes non-additive chore valuations. Working with doubly monotone valuations under the two-sided normalisation, it separates the allocation and subsidy steps that are handled jointly in [BDN⁺20]: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n - 1$ per agent and $n(n - 1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, this yields the stated subsidy guarantee throughout that class. This result, therefore, provides the natural benchmark against which we compare our subsidy bound.

A complementary approach dispenses with subsidies and instead considers approximate notions of envy-freeness. Bu et al. [BSY23] show that EFX allocations always exist and can be computed in polynomial time for dichotomous goods valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. These results provide two complementary guarantees on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. This paper establishes the first branch for chores.

The chores side. Subsidies have also been studied in the allocation of indivisible chores, although much of this literature concerns fairness notions other than envy-freeness. Subsidies have, in particular, been studied for achieving proportionality under additive costs. For agents with equal entitlements, tight subsidy bounds are known: the optimal total subsidy is $n/4$ when n is even and $(n^2 - 1)/(4n)$ when n is odd [WXZ25]. In the more general weighted setting, a total subsidy of $n/3 - 1/6$ is known to suffice [WZ24], and Garg et al. [GSW25] obtain this guarantee together with Pareto optimality. These results concern

proportionality rather than envy-freeness, and therefore do not directly bound the subsidy required in our setting.

Turning to approximate fairness without subsidies, there is a direct chore analogue of [BSY23]. Tao et al. [TWYZ25] study chores with binary marginals and show that for binary *additive* costs, an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half pays 0 or 1 per agent and obtain exact envy-freeness, is what this paper establishes. The restriction to additive costs there cannot simply be dropped: Lin et al. [LLTZ26] exhibit instances with 18 agents and 53 chores, one with binary XOS costs and one with binary supermodular costs, in which no complete EFX allocation exists at all; the binary submodular case remains open. Thus, complete EFX allocations need not exist throughout the binary-marginal chore domain, whereas the allocation constructed below is always EF1, and becomes exactly envy-free once at most one unit of subsidy per agent is paid. Exact envy-freeness without subsidies is not always attainable; indeed, Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Recent work has also considered subsidies for additive instances involving both goods and chores. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN⁺20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $|v_i(g)| \leq 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n - 1$ in total. In particular, this establishes the optimal subsidy bound for additive chore instances.

This result does not subsume the present setting, for the same reason that [BKNS22] is not a corollary of [BDN⁺20]: dichotomous valuations need not be additive, and additive valuations need not be dichotomous. The relationship among these results can be summarised as follows:

	goods	chores
additive	[BDN ⁺ 20]	[LMS26]
dichotomous	[BKNS22]	this paper

each of the four yielding a per-agent subsidy bound of one and a total subsidy bound of $n - 1$. Note also that [LMS26] does not subsume [KMS⁺25] for the valuation class considered here: for valuations that are dichotomous but not additive, the $n - 1$ per agent of [KMS⁺25] was the best previously known bound, and the present result improves this by a linear factor.

To the best of our knowledge, envy-freeness with subsidies has not previously been studied for chores under dichotomous costs.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation

$v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i \in N} \rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An allocation $\mathcal{A} = (A_1, \dots, A_n)$ is an ordered tuple of pairwise disjoint bundles, with A_i assigned to agent i . The allocation is *complete* if $\bigcup_{i \in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \dots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\text{SW}(\mathcal{A}) = \sum_{i \in N} v_i(A_i)$.

Definition 3 (Envy-Freeness). *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ is envy-free (EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution). *An allocation $\mathcal{A} = (A_1, \dots, A_n)$ and a subsidy vector $p = (p_1, \dots, p_n) \in \mathbb{R}_+^n$ constitute an envy-free solution $(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}_+^n$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $M(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will also use the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS⁺25], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

Definition 5 (EF1, two-sided form). *An allocation $\mathcal{A}$ is envy-free up to one item (EF1) iff for all $i, j \in N$ there exists $X \subseteq A_i \cup A_j$ with $|X| \leq 1$ such that $v_i(A_i \setminus X) \geq v_i(A_j \setminus X)$.*

Every instance in this paper is an all-chore instance, and there Definition 5 takes a simpler form: the item removed can always be taken from the envious agent's own bundle. We record that specialisation, since it is the form the results below establish.

Definition 6 (EF1 for chores). *A complete allocation $\mathcal{A}$ of chores is EF1 iff for all $i, j \in N$, either i does not envy j , that is $c_i(A_i) \leq c_i(A_j)$, or there is a chore $e \in A_i$ with*

$$c_i(A_i \setminus \{e\}) \leq c_i(A_j).$$

Definition 6 is exactly Definition 5 on an all-chore instance. Taking $X = \emptyset$ in the first case and $X = \{e\} \subseteq A_i$ in the second gives one direction; for the converse, a removed item lying in A_j only lowers $c_i(A_j)$'s counterpart on the wrong side of the inequality, so nothing is gained by removing from the envied bundle when every item is a chore.

2.1 Negative Dichotomous Valuations

Definition 7 (Negative Dichotomous Valuation). *A valuation $v_i : 2^M \rightarrow \mathbb{R}$ with $v_i(\emptyset) = 0$ is negative dichotomous iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \quad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \quad \text{so} \quad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}.$$

Definition 8 (Dichotomous cost function). *A set function $c : 2^M \rightarrow \mathbb{R}$ is dichotomous if $c(\emptyset) = 0$ and every marginal $c(S \cup \{g\}) - c(S)$ lies in $\{0, 1\}$.*

Monotonicity is not a separate hypothesis: marginals in $\{0, 1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j , read as “my own burden net of my compensation is no worse than what I perceive of yours”. We use whichever of v_i and c_i is clearer locally and always state which.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) := v_i(A_j) - v_i(A_i) = c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j . The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j) \in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i , the empty path of weight 0 included.

The following two results underpin much of the literature on subsidies for envy-freeness. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 9 ([HS19]). *For any allocation $\mathcal{A} = (A_1, \dots, A_n)$, the following are equivalent.*

- (i) $\mathcal{A}$ is envy-freeable.
- (ii) $\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N , $\sum_{i \in N} v_i(A_i) \geq \sum_{i \in N} v_i(A_{\sigma(i)})$.
- (iii) The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.

Theorem 10 ([HS19]). For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p_i^* := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p_i^* \leq p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.

Condition (ii) of Theorem 9 has the standing consequence that an envy-freeable allocation can always be obtained from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. The central question is therefore not the existence of an envy-freeable allocation, but the amount of subsidy guaranteed by Theorem 10.

Specialising in our model gives the following, which we use throughout.

Observation 11 (Integrality). Let $\{v_i\}_{i \in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\leq 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i, j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}_{\geq 0}^n$.

Proof Fix i and $S = \{g_1, \dots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 7; hence $v_i(S) \in \mathbb{Z}$ and $-k \leq v_i(S) \leq 0$. Consequently each arc weight $w_{\mathcal{A}}(i, j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 10, $p_i^* = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i , and it is non-negative because the empty path has weight 0. $\square$

Observation 11 is what makes the target statement — a subsidy vector in $\{0, 1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p_i^* \leq 1$ for every i is the same as showing $p^* \in \{0, 1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

The following observation further simplifies the target guarantee.

Proposition 12 (Some agent is always unpaid). For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p_h^* = 0$. Consequently, if $p^* \in \{0, 1\}^n$ then $\sum_{i \in N} p_i^* \leq n - 1$.

Proof By Theorem 9(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h , and P followed by Q is a walk from P 's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n - 1$ coordinates of p^* can equal 1. $\square$

Proposition 12 justifies the word “hence” in the target statement of Section 1: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. For the pointwise-minimal subsidy the target is therefore the single-clause statement

$$\exists \mathcal{A} : \max_{i \in N} \ell_{\mathcal{A}}(i) \leq 1.$$

3 Unit Subsidies for Negative Dichotomous Valuations

This section proves the main theorem: for every number of agents n and every negative dichotomous instance there is an allocation admitting an envy-free solution with a subsidy of 0 or 1 per agent, at most $n - 1$ in total, computable in polynomial time. Beyond Definition 8 nothing is assumed of the cost functions — not additivity, not submodularity, not any relation between different agents’ cost functions — and the number of chores is unrestricted.

The argument takes as input an envy-free *partial* allocation produced by the algorithm of Tao, Wu, Yu and Zhou [TWYZ25], and completes it. That algorithm, its Theorem 5.1, and the three update rules recorded in Section 3.1 are due to [TWYZ25] and are restated here for reference; the completion of Section 3.2 onwards, and every lemma and theorem after Lemma 15, are ours.

We work in cost form throughout, so an envy-free solution (Definition 4) is the requirement

$$c_i(A_i) - p_i \leq c_i(A_j) - p_j \quad \text{for all } i, j \in N, \quad (1)$$

and a partial allocation is envy-free (Definition 3) when

$$c_i(A_i) \leq c_i(A_j) \quad \text{for all } i, j \in N. \quad (2)$$

For $S \subseteq M$ and $e \in M \setminus S$ write $c_i(e \mid S) := c_i(S \cup \{e\}) - c_i(S) \in \{0, 1\}$ for a marginal, and for a partial allocation $\mathcal{A}$ write $R(\mathcal{A}) := M \setminus \bigcup_{i \in N} A_i$ for its *residue*.

Observation 13. *For every $i \in N$ and all $S, T \subseteq M$,*

$$c_i(S) < c_i(T) \implies c_i(S) \leq c_i(T) - 1.$$

Moreover $c_i(S) \leq c_i(T)$ whenever $S \subseteq T$.

Proof By Observation 11 every c_i is integer valued, so a strict inequality between two of its values is a gap of at least 1. For monotonicity, enumerate $T \setminus S = \{f_1, \dots, f_q\}$ and telescope: $c_i(T) - c_i(S) = \sum_{u=1}^q c_i(f_u \mid S \cup \{f_1, \dots, f_{u-1}\})$, a sum of terms in $\{0, 1\}$ by Definition 8, hence non-negative. $\square$

3.1 The terminal state of the partial-allocation algorithm

Theorem 14 ([TWYZ25], Theorem 5.1). *For every n and all dichotomous cost functions $c_1, \dots, c_n$ there is a polynomial-time algorithm returning a partial allocation X that is envy-free in the sense of (2) and satisfies $|R(X)| \leq n - 1$.*

The proof below uses not only the statement of Theorem 14 but the state in which the algorithm of [TWYZ25] halts, so we record that algorithm. It maintains a partial allocation $X = (X_1, \dots, X_n)$, envy-free throughout, together with the *equality graph*

$$G = (N, E), \quad (i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j), \quad (3)$$

and repeats the following three rules in the stated order, halting only when none applies.

- (R1) If some $e \in R(X)$ and some $i \in N$ satisfy $c_i(e \mid X_i) = 0$, assign e to i .
- (R2) If some $e \in R(X)$ and some arc $(i, j) \in E$ lying on a directed cycle of G satisfy $c_i(e \mid X_j) = 0$, rotate the bundles along that cycle and assign e to i .
- (R3) Otherwise select a strongly connected component S of G with no arc of E leaving S . If $|R(X)| \geq |S|$, assign one residual chore to each agent of S ; otherwise halt.

The loop runs while $R(X) \neq \emptyset$, so the only exit with $R(X) \neq \emptyset$ is the final clause of (R3).

For the remainder of this section fix the following data: let X be the partial allocation at which the algorithm halts, let $R := R(X)$ and $r := |R|$, and, when $r \geq 1$, let S be the component selected by (R3) at the halting step, with G and E as in (3) evaluated at that state.

Lemma 15. *Suppose $r \geq 1$. Then*

- (a) $r < |S|$;
- (b) $c_i(e \mid X_i) = 1$ for every $i \in N$ and every $e \in R$;
- (c) $c_i(e \mid X_j) = 1$ for every arc $(i, j) \in E$ with $i, j \in S$ and every $e \in R$;
- (d) $c_i(X_i) < c_i(X_j)$ for every $i \in S$ and every $j \in N \setminus S$.

Proof Since $r \geq 1$, the algorithm halted at the final clause of (R3); hence neither (R1) nor (R2) applied at the halting state, and $|R| < |S|$, which is (a).

(b) If $c_i(e \mid X_i) = 0$ for some $i \in N$ and some $e \in R$, rule (R1) would apply. Marginals lie in $\{0, 1\}$ by Definition 8, so the value is 1.

(c) Let $(i, j) \in E$ with $i, j \in S$. Since S is strongly connected there is a directed path in G from j to i ; taking one with no repeated vertex and appending the arc (i, j) yields a directed cycle of G through (i, j) . If $c_i(e \mid X_j) = 0$ for some $e \in R$, rule (R2) would apply to that arc and that cycle. Hence the marginal is not 0, so it is 1.

(d) Let $i \in S$ and $j \in N \setminus S$. No arc of E leaves S , so $(i, j) \notin E$, i.e. $c_i(X_i) \neq c_i(X_j)$ by (3). Envy-freeness (2) gives $c_i(X_i) \leq c_i(X_j)$, so the inequality is strict. $\square$

3.2 The completion

Assume $r \geq 1$ and write $R = \{e_1, \dots, e_r\}$. By Lemma 15(a), $r < |S|$, so we may fix r distinct agents

$$T = \{t_1, \dots, t_r\} \subseteq S,$$

chosen arbitrarily. Define $\mathcal{A} = (A_1, \dots, A_n)$ by

$$A_i := \begin{cases} X_{t_k} \cup \{e_k\}, & i = t_k \ (1 \leq k \leq r), \\ X_i, & i \in N \setminus T. \end{cases} \quad (4)$$

The t_k are distinct and each chore of R is used exactly once, so the bundles of $\mathcal{A}$ are pairwise disjoint and cover M : $\mathcal{A}$ is a complete allocation. Members of T are called *recipients*.

Definition 16 (Subsidy set). *Let $P \subseteq N$ be the smallest set satisfying*

(P1) $T \subseteq P$; and

(P2) *if $i \in N \setminus S$ and $(i, j) \in E$ for some $j \in P$, then $i \in P$.*

Equivalently, P is obtained by initialising $P := T$ and repeatedly adjoining any $i \in N \setminus S$ having an arc of E into the current set, until no further agent can be adjoined; the process terminates because N is finite and P only grows. Define

$$p_i := \begin{cases} 1, & i \in P, \\ 0, & i \in N \setminus P. \end{cases} \quad (5)$$

The complete procedure is summarised in Algorithm 1. The partial-allocation phase in Line 2 is Algorithm 3 of [TWYZ25]; our contribution is the completion and subsidy construction in the subsequent lines.

Algorithm 1 EF1 allocation with at most one unit of subsidy per agent for negative dichotomous valuations

Require: Agents $N = [n]$, chores M , and value-oracle access to negative dichotomous valuations $\{v_i\}_{i \in N}$.

Ensure: A complete allocation $\mathcal{A}$ that is EF1, together with a subsidy vector $p \in \{0, 1\}^n$ such that $(\mathcal{A}, p)$ is envy-free and $\sum_{i \in N} p_i \leq n - 1$.

```

1: Set  $c_i = -v_i$  for every  $i \in N$ .
2: Run Algorithm 3 of [TWYZ25] on  $\{c_i\}_{i \in N}$ .
3: Let  $X = (X_1, \dots, X_n)$  be its terminal partial allocation and let  $R = M \setminus \bigcup_{i \in N} X_i$ .
4: if  $R = \emptyset$  then
5:   return  $(X, 0)$ .
6: Let  $S$  be the tail strongly connected component selected at the terminating application
   of Rule (R3).
7: Write  $R = \{e_1, \dots, e_r\}$ .
8: Choose arbitrary distinct agents  $T = \{t_1, \dots, t_r\} \subseteq S$ .  $\triangleright r < |S|$  by Lemma 15(a)
9: for  $k = 1, \dots, r$  do
10:    $A_{t_k} \leftarrow X_{t_k} \cup \{e_k\}$ .
11: for  $i \in N \setminus T$  do
12:    $A_i \leftarrow X_i$ .
13: Construct the equality graph  $G = (N, E)$  on the agents, where
       $(i, j) \in E \iff i \neq j \text{ and } c_i(X_i) = c_i(X_j)$ .
14: Initialise  $P \leftarrow T$ .
15: while there exist  $i \in N \setminus S$  and  $j \in P$  with  $(i, j) \in E$  and  $i \notin P$  do
16:    $P \leftarrow P \cup \{i\}$ .
17: for  $i \in N$  do
18:    $p_i \leftarrow \mathbf{1}[i \in P]$ .
19:  $\mathcal{A} \leftarrow (A_1, \dots, A_n)$ .
20: return  $(\mathcal{A}, p)$ .
```

Lemma 17. $P \setminus T \subseteq N \setminus S$: every agent of P that is not a recipient lies outside S .

Proof Let $P_0 := T$ and let P_{u+1} be P_u together with every $i \in N \setminus S$ having an arc of E into P_u , so that $P = \bigcup_{u \geq 0} P_u$. We show $P_u \setminus T \subseteq N \setminus S$ by induction on u . For $u = 0$ the set $P_0 \setminus T$ is empty. For the step, every agent adjoined in passing from P_u to P_{u+1} lies in $N \setminus S$ by construction. $\square$

Lemma 18. $p \in \{0, 1\}^n$ and $\sum_{i \in N} p_i = |P| \leq n - 1$.

Proof That $p \in \{0, 1\}^n$ with $\sum_i p_i = |P|$ is immediate from (5). By Lemma 17, $P \subseteq T \cup (N \setminus S)$, and this union is disjoint because $T \subseteq S$; hence $|P| \leq |T| + |N \setminus S| = r + n - |S|$. By Lemma 15(a), $r \leq |S| - 1$, so $|P| \leq n - 1$. $\square$

Lemma 19. Let $i \in S$, let $t \in T$ and let $e \in R$. Then

$$c_i(X_t \cup \{e\}) \geq c_i(X_i) + 1.$$

Proof Note $t \in T \subseteq S$. Envy-freeness (2) gives $c_i(X_i) \leq c_i(X_t)$.

If $c_i(X_i) < c_i(X_t)$, then $c_i(X_t) \geq c_i(X_i) + 1$ by Observation 13, and $c_i(X_t \cup \{e\}) \geq c_i(X_t)$ by monotonicity, again Observation 13.

Otherwise $c_i(X_i) = c_i(X_t)$. If $i = t$ then $c_i(X_t \cup \{e\}) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1$ by Lemma 15(b). If $i \neq t$ then $(i, t) \in E$ by (3), with $i, t \in S$, so $c_i(e \mid X_t) = 1$ by Lemma 15(c) and therefore $c_i(X_t \cup \{e\}) = c_i(X_t) + 1 = c_i(X_i) + 1$. $\square$

3.3 Envy-freeness of the completion

Theorem 20. Let X, R, S be as in Section 3.1 with $r \geq 1$, let $\mathcal{A}$ be as in (4) and let p be as in (5). Then $(\mathcal{A}, p)$ is an envy-free solution.

Proof Fix $i, j \in N$; we verify (1), that is $c_i(A_i) - p_i \leq c_i(A_j) - p_j$. Throughout, $X_j \subseteq A_j$ for every j by (4), so by monotonicity (Observation 13) and (2),

$$c_i(A_j) \geq c_i(X_j) \geq c_i(X_i). \quad (6)$$

Note also that $T \subseteq P$, so $i \notin P$ implies $i \notin T$ and hence $A_i = X_i$; likewise $j \notin P$ implies $A_j = X_j$ and $p_j = 0$. Since N is the disjoint union of $T, P \setminus T$ and $N \setminus P$, the following three cases are exhaustive.

Case 1: $i \in T$. Then $p_i = 1$ and, by Lemma 15(b), $c_i(A_i) = c_i(X_i) + c_i(e \mid X_i) = c_i(X_i) + 1$, so $c_i(A_i) - p_i = c_i(X_i)$. Also $i \in T \subseteq S$.

(1a) $j \in T$. Then $p_j = 1$ and $A_j = X_j \cup \{e\}$ for some $e \in R$, so $c_i(A_j) \geq c_i(X_i) + 1$ by Lemma 19. Hence $c_i(A_j) - p_j \geq c_i(X_i)$.

(1b) $j \in P \setminus T$. Then $p_j = 1$ and $A_j = X_j$. By Lemma 17, $j \notin S$; since $i \in S$, Lemma 15(d) gives $c_i(X_i) < c_i(X_j)$, hence $c_i(X_j) \geq c_i(X_i) + 1$ by Observation 13. Hence $c_i(A_j) - p_j = c_i(X_j) - 1 \geq c_i(X_i)$.

(1c) $j \in N \setminus P$. Then $p_j = 0$ and $A_j = X_j$, so $c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i)$ by (2).

Case 2: $i \in P \setminus T$. Then $p_i = 1$ and $A_i = X_i$, so $c_i(A_i) - p_i = c_i(X_i) - 1$. For every j we have $p_j \leq 1$ and $c_i(A_j) \geq c_i(X_i)$ by (6), hence $c_i(A_j) - p_j \geq c_i(X_i) - 1 = c_i(A_i) - p_i$.

Case 3: $i \in N \setminus P$. Then $p_i = 0$ and $A_i = X_i$, so $c_i(A_i) - p_i = c_i(X_i)$.

(3a) $j \in N \setminus P$. Then $p_j = 0$ and $A_j = X_j$, so $c_i(A_j) - p_j = c_i(X_j) \geq c_i(X_i)$ by (2).

(3b) $j \in T$. Then $p_j = 1$ and $A_j = X_j \cup \{e\}$ for some $e \in R$; we claim $c_i(A_j) \geq c_i(X_i) + 1$. If $i \in S$ this is Lemma 19. If $i \notin S$, suppose $c_i(X_i) = c_i(X_j)$. Then $i \neq j$, since $j \in T \subseteq P$ while $i \notin P$; so $(i, j) \in E$ by (3), and $j \in P$, so (P2) of Definition 16 places $i \in P$ — contradicting $i \notin P$. Hence $c_i(X_i) < c_i(X_j)$, so $c_i(X_j) \geq c_i(X_i) + 1$ by Observation 13, and $c_i(A_j) \geq c_i(X_j)$ by monotonicity. In either case $c_i(A_j) - p_j \geq c_i(X_i)$.

(3c) $j \in P \setminus T$. Then $p_j = 1$ and $A_j = X_j$; we claim $c_i(X_j) \geq c_i(X_i) + 1$. If $i \notin S$, suppose $c_i(X_i) = c_i(X_j)$. Then $i \neq j$, since $j \in P$ while $i \notin P$; so $(i, j) \in E$, and (P2) places $i \in P$, a contradiction. Hence $c_i(X_i) < c_i(X_j)$. If instead $i \in S$, then $j \notin S$ by Lemma 17, and Lemma 15(d) gives $c_i(X_i) < c_i(X_j)$. In either case Observation 13 yields $c_i(X_j) \geq c_i(X_i) + 1$, so $c_i(A_j) - p_j = c_i(X_j) - 1 \geq c_i(X_i)$.

In every sub-case $c_i(A_j) - p_j \geq c_i(X_i) = c_i(A_i) - p_i$ in Cases 1 and 3, and the inequality was shown directly in Case 2. $\square$

3.4 The EF1 property of the completion

Corollary 21. *The complete allocation $\mathcal{A}$ constructed in Section 3.2 is EF1 in the sense of Definition 6, hence also in the sense of Definition 5. More precisely, every non-recipient $i \in N \setminus T$ is envy-free in $\mathcal{A}$, while for every recipient $t_k \in T$, removing the residual chore e_k assigned to t_k eliminates all of t_k 's envy.*

Proof Recall that the partial allocation $X = (X_1, \dots, X_n)$ is envy-free, and that $X_j \subseteq A_j$ for every $j \in N$. First, let $i \in N \setminus T$. Then $A_i = X_i$. Hence, for every $j \in N$, envy-freeness of X and monotonicity of c_i give

$$c_i(A_i) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j).$$

Thus i does not envy any agent in $\mathcal{A}$. Now let $i = t_k \in T$. By construction,

$$A_i = X_i \cup \{e_k\},$$

and hence

$$A_i \setminus \{e_k\} = X_i.$$

Therefore, for every $j \in N$,

$$c_i(A_i \setminus \{e_k\}) = c_i(X_i) \leq c_i(X_j) \leq c_i(A_j),$$

where the first inequality follows from envy-freeness of X and the second from monotonicity.

Thus, for every recipient, removing the single newly assigned chore from her own bundle eliminates her envy towards every other agent. Consequently, $\mathcal{A}$ is EF1. $\square$

3.5 The main theorem

Theorem 22. *For every n and every instance $\langle N, M, \{v_i\}_{i \in N} \rangle$ with negative dichotomous valuations there exist a complete allocation $\mathcal{A}$ and a subsidy vector $p \in \{0, 1\}^n$ with $\sum_{i \in N} p_i \leq n - 1$ such that $(\mathcal{A}, p)$ is an envy-free solution. Moreover, the allocation $\mathcal{A}$ is EF1. Such an allocation and subsidy vector can be computed in polynomial time given value-oracle access.*

Proof Pass to cost form, $c_i = -v_i$, which is dichotomous in the sense of Definition 8. Run Algorithm 3 of [TWYZ25] and retain its terminal partial allocation X , the residue R with $r = |R|$. If $r \geq 1$, retain also the tail strongly connected component S selected at the terminating application of rule (R3), all as fixed in Section 3.1. The statement of Theorem 14 alone does not supply S .

If $r = 0$ then X is a complete allocation satisfying (2), so $(X, \mathbf{0})$ is an envy-free solution with $\sum_i p_i = 0 \leq n - 1$. Since every envy-free allocation is EF1, X is also EF1.

If $r \geq 1$, construct T , $\mathcal{A}$ and p as in Section 3.2. Then $\mathcal{A}$ is a complete allocation, Theorem 20 shows $(\mathcal{A}, p)$ is an envy-free solution, and Lemma 18 gives $p \in \{0, 1\}^n$ with $\sum_i p_i \leq n - 1$. The allocation $\mathcal{A}$ is EF1 by Corollary 21.

For the running time, Algorithm 3 of [TWYZ25] is polynomial and yields X , R and the tail component S of its terminating (R3) step if $r \geq 1$. Constructing $\mathcal{A}$ from (4) takes $r \leq n - 1$ assignments. The graph G of (3) has n vertices and is determined by the n^2 values $c_i(X_j)$, one oracle call each. Computing P is a backward reachability search in G from T restricted to vertices outside S , which takes $O(n^2)$ time. Hence the work after Theorem 14 is polynomial, and so is the whole procedure. $\square$

By Example 1 the bound of Theorem 22 is tight: there are instances on which every complete allocation requires total subsidy at least $n - 1$.

Remark 23. Theorem 22 uses Theorem 14 together with the four properties of its halting state recorded in Lemma 15, which are read off the rules (R1)–(R3) of [TWYZ25] restated in Section 3.1; the statement of Theorem 14 alone does not supply them. Envy-freeness is verified directly from (1) in Theorem 20, so the argument does not invoke Theorem 9 or Theorem 10, and the subsidy exhibited need not be the pointwise-minimal one. Part (a) of Lemma 15 is used only in Lemma 18: the conclusion of Theorem 20 holds whenever $r \leq |S|$.

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